MILITARY SPECIFICATION SHEETS

CHAFF PACKAGE (TYPE RR-177/AL)

This specification is approved for use by the Naval Air System Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the chaff package described herein shall consist of this document and the latest issue of Specification MIL-C-85002.

APPLICABLE DOCUMENTS:

Military:

MIL-R-60346 — Roving, Glass, Fibrous

REQUIREMENTS:

Item description: Aluminum foil or alumunium coated glass fiber dipoles and inner sleeves containing alumnimum spiral dipoles.

Detail requirements:

Components: Dipoles and non-chaff components conforming to DL 1050AS150 and requirements herein.

Inner sleeves: For dipole cuts 1 and 2, inner sleeves shall be constructed in accordance with DL 1050AS150.

Chaff dipoles: Cut numbers, lengths, and quantities of chaff dipoles shall be as specified in table I.

Dipole material: Shall consist of aluminum coated glass monofilaments or alumunum foil.

Aluminum coated glass monofilaments: 3 mil. diameter (nominal). Quantities required (see table I).

Glass monofilament: Shall be drawn from Type E glass marbles made from a composition similar to that specified in MIL-R-60346.

Diameter: 2.95 mil (maximum).

FSC 5865

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TABLE I. Chaff dipole material, dimensions and quantity

Dipole Cut No.	Chaff 1/ Material	Dipole Length (inches)	Dipole Width/ Thickness (± 10%)	No. of Dipoles Per Chaff Cut (+10, -5%) 1/	No. of Cuts Per Package	No. of Dipoles Per Package
1	Aluminum foil	27.00 ± 5%	.125 in./ .0018 in.	2/ 48	2/ 5	240
2	Aluminum foil	13.00 ± 5%	.125 in./ .0018 in.	2/ 98	2/ 5	490
1/ 3	V-Bend foil	6.00 ± 5%	.016 in./ .001 in.	12,500	1	12,500
1/ 4	V-Bend foil	6.50 ± 5%	.016 in./ .001 in.	12,500	1	12,500
1/ 3	3 mil. monofilament	6.00 ± 5%	.003 in. dia.	40,000	1	40,000
1/ 4	3 mil monofilament	6.50 ± 5%	.003 in. dia.	40,000	1	40,000

1/ Alternate materials are shown; aluminum foil or coated glass (see "Dipole Material"). The number of dipoles per chaff cut and package for cut numbers 3 and 4 is reference information.

2/ For quantity, tolerances and alternate count of cuts 1 and 2, see "Winding" and "Chaff Package".

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Aluminum coating: Coating material shall be 99 percent or purer aluminum. Coating shall cover length continuous wound circumference of glass monofilament at no less than .06 mil. thickness.

Aluminum foil dipoles: Dimensions and numbers of aluminum foil dipoles as specified in table I. Dipoles shall be constructed of plain finish hard aluminum foil of 3005 or 5052 alloy. Multiwebb cutting (multiple layers at one cut), if used, shall not result in metal smear, end welding, or any dipole adhesion.

Coating: Glass monofilament and V-bend aluminum foil material, cuts 3 and 4 shall be slipcoated in accordance with MIL-C-85002. Aluminum foil dipoles, cuts 1 and 2, shall not be coated.

Spiral wound dipole assemblies: Shall consist of inner case containing aluminum foil dipoles wound spirally on a core as specified in DL 1050AS150.

Core: Plastic and shall conform to DL 1050AS150.

Windings: Each spiral roll shall be fabricated by consecutively winding 48 +5, -3 Cut 1 or 98 +10, -5 Cut 2 dipoles of foil onto a 1.250 ± .025 inch diameter core. The outside diameter of each spiral roll shall be 2.312 ± .1156 inches. An alternate winding shall consist of 24 +3, -2 cut 1 dipoles and 49 +5, -3 cut 2 dipoles wound one for two alternately on the same core in the same manner. Dipoles shall be wound to fit inner sleeve and will not unwind or tangle during storage, but will readily deploy upon dispensing. Free end of outer dipole shall be taped to the inner sleeve as specified in DL 1050AS150.

Weight, chaff only: 224 grams gross weight (minimum).

Chaff package: Shall consist of an outer sleeve containing 10 spiral wound dipole assemblies (5 each of cut 1 and cut 2) 2 cuts of aluminum foil or aluminum coated glass dipoles and vibration protectors as specified in 1050AS150.

Tape marking: Red ink in accordance with DL 1050AS150.

Assembly:

Chaff package: Ten spiral wound dipole assemblies and 2 groups of straight dipoles shall be assembled in the chaff package (outer sleeve) with non-chaff components as specified in DL 1050AS150. The package shall separate the individual spirals and prevent unwinding prior to dispensing, but shall allow free and total dispersion upon deployment.

Vibration protectors: Conform to DL 1050AS150.

Dimensions: Are permitted to vary to provide a tight assembled package.

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Inner sleeves: Conform to DL 1050AS150.

Wrapping of dipoles: Dipole cuts 3 and 4 (see table I) may be wrapped using Type I, Grade A paper as specified in UU-P-268G. If the wrap is greater than 1 1/2 turns, slit paper longitudinally to facilitate deployment of the dipoles. Chaff or wrap appearance shall not be degraded by slitting process.

Weight, chaff package: 275 grams gross weight (minimum).

Quality assurance provisions:

Radar requirements: UHF radar with 1.5-2 degrees beamwidth horizontally polarized antenna.

Frequency: 435 ± 21.75 MHz

PRF: 320 ± 32 pps

Pulsewidth: 1.0 us

Radar cross-section test procedures:

Aircraft operation: In accordance with figure 3 of MIL-C-85002 and flight profile data of table II.

TABLE II. Flight profile test data.

Flight requirement	Value
Aspect Angle	5° - 20°
Aircraft	180 kts. TAS
Altitude	10,000 feet
Dispense Command	35,000 yards
Standby Command	36,000 yards

Data gathering: For acceptable data in gates 12 through 15, range in yards divided by sine of aspect angle shall be equal to or greater than 211,053.

Standby command: See table II.

Dispense command: See table II.

Data from gates 1 through 15: Spaced 1,000 nanoseconds in time.

Full scale value: 20 m^2.

Data processing:

Radar resolution cables: 1000 to 6500 feet behind the aircraft.

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Data selection:

Running average: 400 consecutive pulse-to-pulse RCS values.

Correction factors:

Aspect angle: Use ℓ/D ratio of 10 and dipole density of 20 dipoles per meter squared.

Antenna gain: Antenna pattern as specified in MIL-STD-2071 or fourth order polynominal.

Fourth order polynonimal for UHF radar: $.0249023X^4 + .168213X^3 + .66C504X^2 + .0185547X + .999268$ (where X = angle off boresite in degrees).

Radar cross-section (RCS): measured 1000 through 6500 feet behind the aircraft shall meet or exceed 5 square meters per gate.

Average of 10 or more units: Must equal or exceed 8 square meters per gate.

Preparing Activity
Navy-AS
(Project 5865-N030-4)

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